

FINLEY JOHN NEILSON

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Full-stack and graphics developer, currently an Honours student at the University of Auckland with a paper accepted at OzCHI 2026. I enjoy building maintainable systems with intuitive user experiences.

EDUCATION

University of Auckland

Bachelor of Science (Honours), Computer Science | GPA: 8.7 / 9 (A+)

March 2026 – November 2026

Auckland, NZ

Victoria University of Wellington

Bachelor of Science, Computer Science and Philosophy

February 2021 – November 2025

Wellington, NZ

EXPERIENCE

Software Developer

WDCC – Web Development & Consulting Club

April 2026 – Present

Auckland, NZ

- Developing the UoA Linux User Group website (**120+ users**) in a **12-person team** with **Next.js**, **TypeScript**, **Tailwind CSS**, **PostgreSQL** and **Prisma**, to be self-hosted on a Raspberry Pi upon release.
- Built a member registration system that remains **fully functional with JavaScript disabled**, per client requirements.
- Designed the **Prisma schema** for the blog and added **server-side validation** to registration.

Computer Science Tutor

COMP 112, Victoria University of Wellington

February 2022 – July 2022

Wellington, NZ

- Helped students in labs and marked assignments for a **500+ student** programming course.
- Received positive feedback for effective teaching methods and clear communication.

Chef

Arcimboldi

October 2021 – October 2024

Seatoun, Wellington

- Cooked to order as part of a tight kitchen team under time pressure for 3 years.

PROJECTS

GeoScratch (Honours Research) | React, Three.js, Blockly, WebGL, Supabase/PostgreSQL

[GitHub](#)

- Co-authored *GeoScratch – A Block Based Language for Geometry Education* on the earlier system, **accepted to OzCHI 2026**, to be presented to **200+** attendees.
- Building a **Scratch-style block language** with **real-time 3D visualisation** for teaching geometry and linear algebra.
- Applying **perception research** to 3D rendering: GPU halos, CIEDE2000-optimised colours and label decluttering.
- Designing animation as explanation rather than decoration, revealing a construction in the order its blocks build it.
- Instrumenting the tool for user studies: a **Supabase** backend with anonymous accounts and per-attempt logging.

Miise | Flutter, Dart, Firebase, GraphQL, PostgreSQL

- Leading a team of 6 developers** building a **Letterboxd-style** social app for recipes.
- Cook mode with **gesture-based step navigation**, so you can move through a recipe with messy hands.
- LLM-powered recipe import**: paste any recipe link and get a structured recipe ready to edit and cook.
- Firebase** backend serving recipes over **GraphQL** from a **PostgreSQL** database.

TerrainPlus | C++, OpenGL, GLSL, ImGui

[GitHub](#)

- Developed a **procedural terrain generation system** using fractal Perlin noise functions.
- Implemented a **thermal erosion simulation** to improve terrain realism through iterative height redistribution.
- Built a **shader-based** terrain texturing system blending sand, grass, stone, and snow textures based on elevation.

Shape Transform | React, TypeScript, Three.js, Node.js, Express, MongoDB

[GitHub](#)

- Largest contributor** to a full-stack **3D spatial-reasoning puzzle game**.
- Developed the core **Three.js** scene, using **quaternions** to rotate pieces about world axes.
- Moved rendering from server to client, **removing image generation** and animation lag on API calls.

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, HTML, CSS, Go, C, C++, Dart, GraphQL | **Tools:** Git, Docker, Neovim, WSL, JetBrains IDEs

Web: MERN Stack (MongoDB, Express, React, Node.js), Next.js, Tailwind CSS | **Mobile:** Flutter

Graphics: Three.js, WebGL, OpenGL, GLSL | **Databases:** PostgreSQL, MongoDB, Firebase, Supabase, Prisma